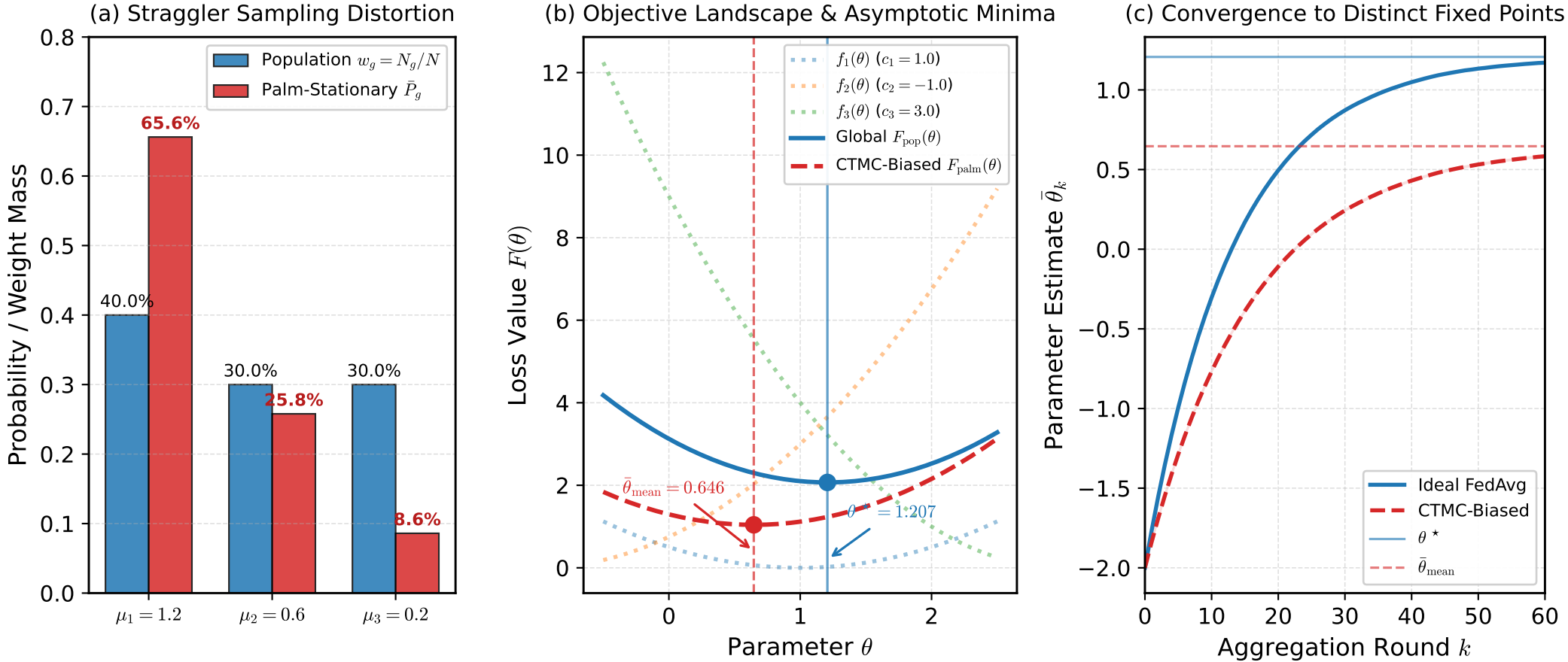
(a) Straggler Sampling Distortion
Population $w_g = N_g/N$
Palm-Stationary $\bar{P}_g$
Probability / Weight Mass
65.6%
40.0%
30.0%
25.8%
30.0%
8.6%
$\mu_1 = 1.2$
$\mu_2 = 0.6$
$\mu_3 = 0.2$
(b) Objective Landscape & Asymptotic Minima
$f_1(\theta)$ ($c_1 = 1.0$)
$f_2(\theta)$ ($c_2 = -1.0$)
$f_3(\theta)$ ($c_3 = 3.0$)
Global $F_{\text{pop}}(\theta)$
CTMC-Biased $F_{\text{palm}}(\theta)$
Loss Value $F(\theta)$
$\bar{\theta}_{\text{mean}} = 0.646$
$\theta^\star = 1.207$
Parameter θ
(c) Convergence to Distinct Fixed Points
Parameter Estimate $\bar{\theta}_k$
Ideal FedAvg
CTMC-Biased
$\theta^\star$
$\bar{\theta}_{\text{mean}}$
Aggregation Round k